

14.0 Summative Assessment: Network Communication Protocols - Lab 17

Overview:

The purpose of this assessment is to showcase your hands-on experience and understanding of network setup, configuration, and troubleshooting.

Instructions:

To complete this summative assessment, choose one of the labs in the competency. Include detailed documentation of your configurations, troubleshooting steps, and results.

Remember:

- Keep detailed notes during each lab to make documenting easier.
- Include screenshots or configuration files to support your documentation.
- Reflect on what you learned in each lab and any challenges you faced.
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Packet Tracer Lab 17 - Configure DHCP on a Wireless Router

1. The lab details and the “no audio” walkthrough YouTube video are both accessible from this page:
<https://herbal-rancher.github.io/network-portfolio/videos/14-17-configure-dhcp-wireless-router/>
2. Here's a link to the “no audio” walkthrough YouTube video:
https://youtu.be/DqZmekPmB08?si=agGN_mICuZVIhaq7
3. From this lab, I was able to complete and understand the following:
 - a. Connect multiple PCs to a wireless router
 - b. Examine default DHCP settings
 - c. Configure a wireless router's IPv4 address
 - d. Modify a DHCP address pool
 - e. Configure clients to obtain IPv4 addresses dynamically
 - f. Verify DHCP-assigned addressing using ipconfig /all command
 - g. Test connectivity using ping

- h. Validate communication between network devices
- 4. This was definitely an easier lab and less stressful because it included step by step instructions; however, I appreciated the detail because it allowed me to become familiar with the steps, environments, successful and unsuccessful attempts at connecting, configuring, verifying and testing.

Objectives

- Connect 3 PCs to a wireless router
- Change the DHCP setting to a specific network range
- Configure the clients to obtain their address via DHCP

Background / Scenario

A home user wants to use a wireless router to connect 3 PCs. All 3 PCs should obtain their address automatically from the wireless router.

Instructions

Part 1: Set up the network topology

- a. Add three generic PCs.
- b. Connect each PC to an Ethernet port to the wireless router using straight-through cables.

Part 2: Observe the default DHCP settings

- a. After the amber lights have turned green, click **PC0**. Click the **Desktop** tab. Select **IP Configuration**. Select **DHCP** to receive an IP address from **DHCP Enabled Router**.

Record the IP address of the default gateway: **192.168.0.1**

- b. Close the **IP Configuration** window.
- c. Open a Web Browser.
- d. Enter the IP address of the default gateway recorded earlier into the URL field. When prompted, enter the username **admin** and password **admin**.
- e. Scroll through the Basic Setup page to view default settings, including the default IP address of the wireless router.
- f. Notice that DHCP is enabled, the starting address of the DHCP range and the range of addresses available to clients.

Part 3: Change the default IP address of the wireless router.

- a. Within the Router IP Settings section, change the IP address to: **192.168.5.1**.
- b. Scroll to the bottom of the page and click **Save Settings**.

- c. If it is done correctly, the web page will display an error message. Close the web browser.
- d. Click **IP Configuration** to renew the assigned IP address. Click **Static**. Click **DHCP** to receive new IP address information from the wireless router.
- e. Open the web browser, enter the IP address **192.168.5.1** in the URL field. When prompted, enter the username **admin** and password **admin**.

Part 4: Change the default DHCP range of addresses.

- a. Notice the DHCP Server Start IP Address is updated to the same network as the Router IP.
- b. Change the Starting IP Address from 192.168.5.100 to **192.168.5.126**.
- c. Change the Maximum Number of Users to **75**.
- d. Scroll to the bottom of the page and click **Save Settings**. Close the web browser.
- e. Click **IP Configuration** to renew the assigned IP address. Click **Static**. Click **DHCP** to receive new IP address information from the wireless router.
- f. Select **Command Prompt**. Enter **ipconfig**.

Record the IP address for PC0: **192.168.5.126**

Part 5: Enable DHCP on the other PCs.

- a. Click **PC1**.
- b. Select **Desktop** tab.
- c. Select IP Configuration.
- d. Click **DHCP**.

Record the IP address for PC1: **192.168.5.127**

- e. Close the configuration window.
- f. Enable DHCP on **PC2** following the steps for PC1.

Part 6: Verify connectivity

- a. Click **PC2** and select the **Desktop** tab.
- b. Select Command Prompt.
- c. Enter **ipconfig** at the prompt to view the IP configuration.
- d. At the prompt, enter **ping 192.168.5.1** to ping the wireless router.
- e. Enter **ping 192.168.5.126** to ping PC0 at the prompt.
- f. At the prompt, enter **ping 192.168.5.127** to ping PC1.
- g. The pings to all devices should be successful.

<https://herbal-rancher.github.io/portfolio/>